The Aisha Manifold:
A Per-Cell Symplectic–Kähler Substrate for Lexical Semantics

Scientific Basis

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Abstract

We construct a dense per-cell symplectic–Kähler manifold over a 50,000-lemma English vocabulary. Each word w carries an algebraically closed 16×16 object from its corpus signature alone: eight part-of-speech channels, a vector symplectic lock $\alpha_w^{(g)} \beta_w^{(g)} = 1$, and a derived shell of Keplerian quantities. The same shell admits a relativistic-quantum reading: the $\alpha\beta$ lock is the angular-momentum invariant of an 8D Hermitian metric, M_w is an antisymmetric torsion (spin), and the pair-torsion $\mathrm{Re}\langle \hat{q}_w, \hat{q}_v \rangle$ plays the role of spin–orbit coupling between adjacent words. This correspondence is what the framework has been moving toward since we swapped the 8D celestial-mechanics picture for the 8D relativistic quantum picture. On top of the three trained axes (M, χ, s) we train two further orthogonal per-word coordinates — valence v and arousal a — that close the long-standing valence-invisibility gap. The same analytical-jump detector used to find orbital pericentre is applied scale-invariantly at three linguistic levels (word $\to$ phrase, phrase $\to$ clause, turn $\to$ discourse); six phrase types and five clause types emerge without supervision. Real English phrases are shown to obey measurable geometric laws: $M(r)$ follows a U-shape within a phrase, χ rises monotonically, valence and arousal arc up in the middle and fall at close. The deployable stack is a two-tier architecture: the physics engine emits a structured template at ~ 0.05 J per query, and a small on-device LM (~ 10 J) fills the surface form. Total per-reply energy is $\sim 100\times$ cheaper than a cloud LLM call of comparable behaviour, with zero data leaving the device, and grammar cleaner than the training corpus (0.02 Harper lints per reply vs 0.12 for corpus humans).

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1 Motivation and overview

Standard distributional semantics encodes meaning as dense neural embeddings learned end-to-end from co-occurrence statistics. The Aisha manifold takes a different route: starting from the same

raw corpus signature, we build a fixed algebraic object per word, where every quantity of interest follows from one trainable scalar per (word, POS) pair via closed-form identities borrowed from the symplectic–Kähler geometry of an analytical Kepler propagator.

The construction has three desirable properties:

1. **Locality.** Each word's state is a small 16×16 matrix; the vocabulary scales linearly in storage and parallelisable in compute. No global matrix appears anywhere.

2. **Algebraic closure.** Once a single per-channel parameter $\alpha_w^{(g)}$ is set (from corpus or by training), every other quantity — velocity, energy, torsion, frequency, eccentricity — is derived without iteration.

3. **Two orthogonal probes.** The same per-cell object supports a spectral-interference scoring layer (sentence as empirical measure on $\mathbb{R}^8$) and a linear-response perturbation layer (sentence as a mass-doubling event on the manifold), which together give a deployable retrieval engine.

2 Per-cell construction

2.1 Vocabulary and per-channel counts

Let $\mathcal{V} = \{w_1, \ldots, w_N\}$ be the vocabulary retained from a corpus pass after tokenisation, lemmatisation, and part-of-speech tagging. Each word carries a $P = 8$-dimensional channel vector of observed counts:

$$m_w \;=\; \big(m_w^{(1)}, \ldots, m_w^{(P)}\big) \in \mathbb{Z}_{\geq 0}^{P}, \tag{1}$$

where channel $g \in \{1, \ldots, 8\}$ groups Universal POS tags as:

channel	role	UD tags
$g = 1$ Noun	content / referent	NOUN, PROPN, NUM
$g = 2$ Verb	content / action	VERB, AUX
$g = 3$ Adj	content / property	ADJ
$g = 4$ Adv	content / modifier	ADV
$g = 5$ Pro	function / pronominal	PRON, DET
$g = 6$ Prep	function / relational	ADP
$g = 7$ Conj	function / connective	CCONJ, SCONJ, PART
$g = 8$ Interj	function / discourse	INTJ

The total count is $m_w = \sum_g m_w^{(g)}$ and the POS membership distribution is $\pi_w^{(g)} = m_w^{(g)}/m_w$, supported on the 8-simplex.

2.2 The anisotropic scale

Per-channel counts are smoothed with a uniform Dirichlet prior of strength ε (default $\varepsilon = 1$), scaled against the channel-wise vocabulary mean, and anchored per word so the geometric mean across

channels is unity:

$$\tilde{m}_w^{(g)} = m_w^{(g)} + \varepsilon \tag{2}$$

$$\bar{m}^{(g)} = \frac{1}{N} \sum_{u \in \mathcal{V}} \tilde{m}_u^{(g)} \tag{3}$$

$$\tilde{\alpha}_w^{(g)} = \tilde{m}_w^{(g)} / \bar{m}^{(g)} \tag{4}$$

$$G_w = \Big(\prod_g \tilde{\alpha}_w^{(g)} \Big)^{1/P} \tag{5}$$

$$\alpha_w^{(g)} = \tilde{\alpha}_w^{(g)} / G_w \tag{6}$$

The final identity enforces, exactly per word,

$$\langle \log \alpha_w \rangle_g \;=\; 0, \tag{7}$$

which fixes a natural gauge: the isotropic cell is $\alpha_w \equiv \mathbf{1}$ and any departure must be reciprocal across channels. This is the information-budget condition associated with treating the Kähler potential as constant per fibre.

2.3 The vector symplectic lock

The per-channel lock is a per-word algebraic identity:

$$\alpha_w^{(g)}\, \beta_w^{(g)} = 1, \qquad \beta_w^{(g)} = 1/\alpha_w^{(g)}, \qquad g = 1, \ldots, 8. \tag{8}$$

Eight independent locks per cell, jointly written as $\mathrm{diag}(\alpha_w)\,\mathrm{diag}(\beta_w) = I_8$.

2.4 Intra-cell channel coupling

Each cell carries a non-negative 8×8 coupling $A_w \in \mathbb{R}_{\geq 0}^{8\times 8}$ assembled from trigram contexts. For each occurrence of w with immediate filtered neighbours of POS g_{prev} and g_{next} we increment $c_w[g_{\mathrm{prev}}, g_{\mathrm{next}}]$. The matrix is then row-stochastic per g:

$$A_w[g,h] = \frac{c_w[g,h]}{\sum_{h'} c_w[g,h'] + \eta}, \qquad \sum_h A_w[g,h] = 1, \tag{9}$$

with η a tiny floor to handle empty rows (which fall back to uniform 1/8). Crucially we do *not* store specific bigram partner words; A_w aggregates over all neighbour identities, keeping each cell local.

2.5 The per-cell symplectic Jacobian

With $B_w \equiv A_w$, the per-cell Jacobian assembles to

$$J_w = \begin{pmatrix} \mathcal{A}_w A_w & \mathcal{B}_w B_w \\ -\mathcal{B}_w B_w & (\mathcal{A}_w A_w)^+ \end{pmatrix} \in \mathbb{R}^{16\times 16}, \tag{10}$$

where $\mathcal{A}_w = \mathrm{diag}(\alpha_w)$, $\mathcal{B}_w = \mathrm{diag}(\beta_w)$, and $(\cdot)^+$ is the Moore–Penrose pseudoinverse (which we use because rare-context rows of A_w may default to uniform, leaving the matrix rank-deficient). The cell-level symplectic form is $\Omega_8 = \left(\begin{smallmatrix} 0 & I_8 \\ -I_8 & 0 \end{smallmatrix}\right)$, satisfying $J_w^\top \Omega_8 J_w = \Omega_8$ exactly when A_w is isotropic and to within finite-context corrections otherwise.

2.6 The Kähler complex coordinate

Each cell additionally carries a complex coordinate $q_w \in \mathbb{C}^8$ seeded by $\left|q_w^{(g)}\right|^2 = \alpha_w^{(g)}$ with $\arg q_w^{(g)} = 0$. The Kähler potential on the lexical substrate, in Fubini–Study form,

$$K(q, \bar{q}) = \ln\Big(1 + \sum_{w,g} w_{w,g}\, e^{\,q_w^{(g)}\overline{q_w^{(g)}}}\Big), \tag{11}$$

serves as the generating function for the information metric and is treated as a conserved quantity per fibre, equivalent to our geometric anchor $\langle \log \alpha_w \rangle_g = 0$.

3 The Keplerian shell

Once $\alpha_w^{(g)}$ is set, every Keplerian quantity follows algebraically. Per-channel gravitational parameter:

$$\mu^{(g)} = \sum_{u \in \mathcal{V}} m_u^{(g)}, \tag{12}$$

in natural units where the gravitational constant is absorbed. Specific energy, semi-major axis, orbital frequency, and hyperbolic torsion follow by direct substitution of the lock:

$$\varepsilon_w^{(g)} = \tfrac{1}{2}(\beta_w^{(g)})^2 - \mu^{(g)}/\alpha_w^{(g)}, \tag{13}$$

$$a_w^{(g)} = -\mu^{(g)}\big/\big(2\varepsilon_w^{(g)}\big), \tag{14}$$

$$\omega_w^{(g)} = \sqrt{\mu^{(g)}\Big/\left|a_w^{(g)}\right|^3}, \tag{15}$$

$$M_w^{(g)} = \sqrt{\big(\alpha_w^{(g)} + \beta_w^{(g)}\big)^2 - 4} = \left|\alpha_w^{(g)} - \beta_w^{(g)}\right|. \tag{16}$$

The last identity, $M = |\alpha - \beta|$, is an algebraic consequence of the lock (substituting $\beta = 1/\alpha$ into the hyperbolic invariant $(\alpha + \beta)^2 - M^2 = 4$).

3.1 Pair-wise torsion

For two words $w \neq v$ we define a scalar pair-wise torsion via a reduced-α construction. Let $\alpha_w^{\text{scalar}} = \pi_w \cdot \alpha_w$ be the POS-weighted channel mean of α_w. Then:

$$\alpha_{w,v} = \frac{\alpha_w^{\text{scalar}}\alpha_v^{\text{scalar}}}{\alpha_w^{\text{scalar}} + \alpha_v^{\text{scalar}}}, \quad \beta_{w,v} = 1/\alpha_{w,v}, \quad M_{w,v} = |\alpha_{w,v} - \beta_{w,v}|\,. \tag{17}$$

3.2 Per-word eccentricity

Eccentricity is computed over the observed bigram neighbours of w:

$$e_w = \Gamma \cdot \frac{M_w^{\max} - M_w^{\min}}{M_w^{\max} + M_w^{\min}}, \qquad \Gamma = 3/16, \tag{18}$$

with $M_w^{\max}, M_w^{\min}$ taken over $\{M_{w,v} : v \in \text{Neighbours}(w)\}$.

3.3 Empirical Keplerian properties

On a 50,000-cell manifold trained on a balanced English pilot (Wikipedia + OpenSubtitles, ≈ 400 M tokens), every cell is algebraically valid:

- Lock residual $\left|\alpha_w^{(g)}\beta_w^{(g)} - 1\right|$ is at the floating-point machine epsilon ($\approx 10^{-16}$).
- Every cell is bound: $\varepsilon_w^{(g)} < 0$ across all $50{,}000 \times 8 = 400{,}000$ (word, channel) entries.
- Per-word eccentricity over 9.4 million observed bigram pairs distributes with median 0.087 (Mars-like), 57% of words below $e = 0.10$ (Saturn / Jupiter / Uranus zone), and $\sim 8\%$ saturated near the ceiling $\Gamma = 0.1875$.

4 Sentence-level structure

4.1 Spectral signature and interference

A sentence Q of $|Q|$ tokens is identified with the empirical measure on $\mathbb{R}^8$,

$$P_Q = \frac{1}{|Q|}\sum_{w\in Q}\delta\big(\,\cdot - \tilde{\boldsymbol{\omega}}_w\big), \qquad \tilde{\boldsymbol{\omega}}_w = \boldsymbol{\pi}_w \odot \boldsymbol{\omega}_w, \tag{19}$$

using the eight-channel effective frequency $\tilde{\omega}_w$. Two sentences interfere by the sliced 1-Wasserstein distance:

$$\widehat{W}_1^{(8)}(Q,R) = \frac{1}{K}\sum_{k=1}^{K} W_1\big(\theta_k^\sharp P_Q,\ \theta_k^\sharp P_R\big), \qquad \theta_k \sim \mathrm{Unif}(S^7), \tag{20}$$

each one-dimensional W_1 being a quantile-matched sorted-L^1 sum.

4.2 Perturbation probe

A complementary scoring layer treats each sentence as an initial-value perturbation on the manifold. Given input sentence S, double the observed counts of every word $w \in S$:

$$\tilde{m}_w^{(g)} \ \mapsto\ 2\cdot\tilde{m}_w^{(g)}, \quad \forall w \in S, \forall g. \tag{21}$$

The α stack is recomputed; the displacement vector

$$\Delta \log \alpha_u^{(g)}(S) = \log \alpha_{u,\mathrm{new}}^{(g)} - \log \alpha_u^{(g)} \tag{22}$$

is an $8N$-vector indexed by (word, channel). Words u sharing S's POS fingerprint experience similar channel-mean shifts. The probe distance between two sentences is the cosine distance on these profiles.

4.3 Per-position context chain

For a sentence $S = (w_1, \ldots, w_n)$ of in-vocab tokens, the context state on the POS simplex propagates via the row-stochastic intra-cell coupling:

$$s_0 = \boldsymbol{\pi}_{w_1}, \qquad s_t = s_{t-1}\cdot A_{w_t}. \tag{23}$$

The trajectory $\{s_t\}_{t=1}^n \in (\Delta^7)^n$ is a sentence-level fingerprint of the grammatical-flow shape, distinct from both the spectral measure and the perturbation profile.

4.4 Three-way fusion

The three signals (interference, probe, chain) are nearly uncorrelated (Pearson $\rho_{\text{probe,chain}} = -0.04$, $\rho_{\text{probe,int}} = 0.17$). Rank-sum fusion with zero hyperparameters:

$$\text{score}(R \mid Q) = \text{rank}_{\text{int}}(R) + \text{rank}_{\text{prb}}(R) + \text{rank}_{\text{chn}}(R), \tag{24}$$

gives the best measured matched-<-random and MRR on Alpaca validation. Adding ratio or complexity signals provides no further improvement (they are correlated with interference).

5 Empirical results

5.1 Manifold construction at scale

property	**value**
vocabulary size N	50,000
tokens in tagged corpus	$\sim$ 400 M
bigram pairs observed	$\sim 10^7$
storage (full manifold)	$\sim$ 210 MB
storage after dropping J	$\sim$ 110 MB
$\langle \log \alpha \rangle$ overall	1.2×10^{-17} (anchor exact)
lock residual max	1.1×10^{-16}
% bound channels	100
median per-word eccentricity	0.087

5.2 Training of α

Only α is trainable; $\beta = 1/\alpha$ follows from the lock and ω from the Kepler map. The objective is a contrastive hinge on sliced-$W_1^{(8)}$ (paper Eq. 10):

$$\mathcal{L} = \mathbb{E}_{(Q,R^+,R^-)} \max\big(0,\, W_1^{(8)}(Q,R^+)^2 - W_1^{(8)}(Q,R^-)^2 + \Delta\big). \tag{25}$$

Hard-negative sampling: R^- drawn from the top-quartile of sentences whose POS-composition vector is most L^2-far from R^+'s.

On Alpaca instruction-pair data (41 k training pairs), training saturates after $\approx$ 3,400 steps (10 epochs, batch 128, cosine LR $5 \times 10^{-2} \to 5 \times 10^{-4}$). Best validation matched-<-random fraction: 0.6735.

5.3 Retrieval benchmarks

Six single methods compared on 295 Alpaca queries $\times$ 10 random negatives each:

method	**matched-<-rand**	**top-1**	**MRR**
probe	0.6342	0.200	0.390
chain	0.5847	0.153	0.343
interference	0.5746	0.180	0.355
ratio	0.5495	0.112	0.303
complexity	0.4956	0.102	0.273
fusion (probe + chain + int)	**0.6397**	0.247	**0.438**

Probe is the best single signal; three-way rank-sum fusion adds marginal but consistent improvement in both retrieval correctness and ranking quality.

5.4 Compression

Probe vectors are 533,600-dim ($\sim$ 2.1 MB each). We investigated three compression strategies:

form	bytes/probe	short	mid	long	all
full	2,134,400	0.679	0.666	0.629	0.642
sparse top-50	**400**	**0.743**	**0.773**	**0.755**	**0.760**
sparse top-100	800	0.721	0.763	0.744	0.749
sparse top-200	1,600	0.714	0.721	0.710	0.714
sparse top-500	4,000	0.700	0.696	0.681	0.686
PCA-256	1,024	0.607	0.540	0.627	0.599

The unexpected finding: **sparse top-**50 **both compresses** 5,300$\times$ **and improves retrieval by** $+12$ **pp on matched-<-random**. PCA, by contrast, destroys short-query discriminability (short-query frac collapses from 0.661 to 0.481 at any dimension between 32 and 512). We attribute the sparse improvement to a denoising effect: the full $\Delta \log \alpha$ vector contains near-zero noise across $\sim$ 533,550 dimensions which dilutes cosine similarity; keeping only the top-50 entries preserves the structural signal.

5.5 Sentence dynamics

Per-position trajectories of KE $= \frac{1}{2}\beta^2$, PE $= \mu/\alpha$, $|\varepsilon|$, $M_r = |\alpha - \beta|$, and $M_\theta = \omega$ along a sentence reveal two distinct families: sentences opening with a determiner or content noun show energy *building* toward the end (positive log(KE/PE) slope across positions), while sentences opening with a pronoun or interjection show energy *decaying* (negative slope). The auxiliary verb "be" acts as a global energy minimum: every occurrence of "is/be/am/are" sits at a deep saddle in KE, PE, and $|\varepsilon|$.

Cumulative torsion $\sum_t M_w^{(g)}$ along a sentence behaves as a "content temperature": two same-length sentences can differ by an order of magnitude in $\sum M_\theta$, with the larger value indicating rare or technical vocabulary.

6 Limitations and open directions

6.1 Valence invisibility — resolved by the (v, a) axes

The early construction could not distinguish "I love you" from "I hate you": both share an identical Pro-Verb-Pro skeleton and the probes respond nearly identically. The resolution documented in §7 is to train two additional per-word axes — valence v_w and arousal a_w — with a logistic contrastive objective on emotion-labelled sentences. The resulting axes carry held-out sentence-level AUC 0.79 (valence) and 0.82 (arousal), and their correlations with the existing (M, χ, s) stack are all $|\rho| < 0.07$: genuinely orthogonal signal. With these two axes in place, the Pro-Verb-Pro skeleton of "I love you" and "I hate you" are separated cleanly by v alone.

Historically we catalogued three forward paths before the (v, a) axes were trained; we retain them for provenance:

(a) Per-word weights w_i in K. Set $w_i \propto m_w^{(g)}/\mu^{(g)}$ instead of uniform. This gives the Kähler metric a data-derived weighting and may sharpen the per-cell information budget without altering the fibre.

(b) K-constant anchor. Replace the per-word geometric anchor $\langle \log \alpha_w \rangle_g = 0$ with the strict condition $K_w = K_{\text{target}}$ via a per-step rescaling. This projects updates onto the Kähler-conserved hypersurface during training.

(c′) Soft lock with independent β. Promote $\log \beta_w^{(g)}$ to an independent trainable parameter and impose $\alpha\beta = 1$ as a soft penalty. At convergence, examine the empirical lock violation distribution; this gives a falsifiable test of whether the data demands an asymmetric phase space.

(c″) Full complex Kähler. Make $q_w^{(g)} \in \mathbb{C}$ trainable in both magnitude and phase, with $K = \ln(1 + \sum w_i e^{|q_i|^2})$ generating a true Hermitian metric on the lexical substrate. Phase becomes meaningful via the Kähler form $\Omega = i\,\partial\bar{\partial}K$, and cosine distance on complex coordinates would carry valence information. This is the path-of-record extension for the next thread of investigation.

6.2 Asymmetric matching

The probe maps a query's response with a systematic ratio bias: matched responses tend to have $\sim$ 3–19$\times$ lower M_r/M_θ ratio than the query. This may reflect a corpus artefact (Alpaca answers are content-dense) or a deeper structural asymmetry between question and answer roles; further measurement on varied datasets is warranted.

6.3 Generation

The Aisha manifold is a retrieval and analysis substrate, not a generative model. An immediate downstream architecture is to use the three-way fusion to retrieve top-k candidate sentences from a pool, and pass them as soft conditioning to a lightweight neural LM (e.g. a 50 M-parameter transformer) for syntactic polishing. The manifold side carries the semantic and structural retrieval signal; the LM side handles only surface-form generation.

6.4 Phone-class deployment

With sparse top-50 probe encoding, the on-device footprint for a 10,000-sentence retrieval pool is approximately 140 MB (manifold + pool features), well within mobile-class storage. Per-query inference is currently $\sim$ 5 s on a single CPU core for a $\sim$ 300-sentence pool; vectorising the sliced-W_1 inner loop should bring this to sub-second.

7 Valence and arousal axes

The three per-word axes used throughout the preceding sections (M, χ, spin) are derived from the POS-channel structure of the manifold and carry amplitude / topology information. They do not separate emotionally signed content because M_w is a magnitude and χ is a neighbour-averaged inner product.

Training data. The conversations corpus (`conversations.csv`: 4,002 turns, 99 conversations, five registers) gives us labelled positive / negative signals. Define E = tokens in emotional $\cup$ heated sentences, N = tokens in casual $\cup$ civilized $\cup$ scientific sentences. A log-odds-ratio probe on the $\sim$ 1,200 content lemmas occurring in both classes with ≥ 5 occurrences gives sentence-level AUC 0.878 against the E/N split — the signal is real before any training.

Training objective. We train one scalar per word, $v_w \in \mathbb{R}$, with a logistic contrastive objective:

$$\hat{y}_S = \sigma\Big(a\,\overline{v}_S + b\Big), \qquad \overline{v}_S = \frac{1}{|S_{\text{content}}|} \sum_{w \in S_{\text{content}}} v_w, \tag{26}$$

with cross-entropy loss against $y_S = \mathbb{1}[S \in E]$ and an L^2 regulariser on v. The anchor $\langle v \rangle_{w \in V_{10k}} = 0$ is enforced per training step. The same objective on a *heated*-only label trains a second scalar a_w (arousal); the two axes are trained independently.

Held-out behaviour.

axis	held-out sentence AUC	Cohen's d
valence v	0.786	1.28
arousal a	0.815	1.40 (heated-vs-civilized)

Orthogonality. Per-word Pearson correlations of the trained axes with the existing shell:

axis pair	M	χ	s	$M{+}\chi{+}s$
v	+0.06	−0.07	+0.02	+0.02
a	−0.03	+0.03	+0.02	+0.04

All correlations $|\rho| < 0.07$: v and a are new, orthogonal coordinates on the lexical manifold, not a rotation of the old ones. In a quantum-mechanical reading they are analogous to the two Cartan charges of an additional SU(2) acting on the lexical fibre; in the Kähler picture they extend the complex coordinate $q_w \in \mathbb{C}^8$ with two real scalars that *cannot* be written as a function of α_w or β_w.

Style centroids. Per-register mean of each axis over the trained manifold reproduces the affective structure of the corpus without further supervision:

	M	χ	s	v	a
casual	+2.06	−2.13	+0.17	−0.30	−0.65
civilized	+2.09	−2.13	+0.16	−0.40	−0.65
emotional	+2.16	−2.30	+0.20	+0.15	−0.55
heated	+2.07	−2.09	+0.23	+0.29	+0.44
scientific	+1.99	−1.97	+0.10	−0.66	−0.82

The five registers are indistinguishable on the (M, χ, s) axes but separate cleanly in the (v, a) plane: heated (+,+), emotional (+,−), scientific (−, −−), civilized / casual (−, −).

8 The analytical-jump hierarchy

The analytical jump — the detector that identifies orbital pericentre as a large step in the per-timestep coordinate trajectory in the 8D celestial-mechanics work — is applied scale-invariantly at three linguistic levels: word → phrase, phrase → clause, turn → discourse. At each level the rule is identical: compute a composite jump score J_k between consecutive units, detect positions where J_k is a local maximum above threshold, and partition the sequence at those positions.

Layer 2: word → phrase. Between consecutive words (w_k, w_{k+1}) we compute four candidate jump measures:

$$\Delta\varepsilon_k = |\varepsilon_{w_{k+1}} - \varepsilon_{w_k}| \qquad \text{(Kepler energy)} \tag{27}$$

$$\Delta e_k = \big|\, |e_{w_{k+1}}| - |e_{w_k}| \,\big| \qquad \text{(torsion eccentricity)} \tag{28}$$

$$\Delta_k^{5D} = \big\| \boldsymbol{\xi}_{w_{k+1}} - \boldsymbol{\xi}_{w_k} \big\|_2, \qquad \boldsymbol{\xi}_w = (M_w, \chi_w, s_w, v_w, a_w) \tag{29}$$

$$\Delta_k^{\text{pair}} = 1 - \text{Re}\,\langle \hat{q}_{w_k}, \hat{q}_{w_{k+1}} \rangle \qquad \text{(pair-torsion drop)} \tag{30}$$

each one z-standardised per sentence; the composite $\tilde{J}_k = \sum_m z(\Delta_k^{(m)})$ drives boundary detection. ^{5D} and e agree on 53% of boundaries (torsion-geometric signal); ^{pair} is nearly independent (Jaccard 0.03–0.07 with the others) and contributes orthogonal information.

On the 4,002-turn corpus, at threshold $\sigma = 1.0$: 2–3 phrases per sentence emerge with median size 3–4 words — matching typical English phrase sizes without supervision. A within-phrase pair-torsion of +0.085 exceeds the across-boundary mean of −0.015 by $\Delta = 0.10$: groups are more tightly bound internally than across their edges, which is the operational definition of a phrase.

Unsupervised KMeans on a 21-dim phrase signature (octant histogram, axis means, coherence, length, relative position) yields six stable phrase types:

- *openers* (8%) — length 1.5, dei-dominated (98%), 60% sentence-initial;
- *loose bridges* (40%) — length 4.8, dei/abs mix, coherence +0.06 (the loosest type);
- *connectives* (16%) — length 2.5, abs-dominated, 75% stopwords;
- *content bodies* (28%) — length 3.1, balanced octants, 54% sentence-final;
- *dense closers* (7%) — length 1.3, rel/coll/cat content, register $v = -1.9, a = -2.4$, coherence +0.86 (tightest type); 66% sentence-final;
- *tag concluders* (1%) — length 1.7, stat octant only, 64% sentence-final.

Layer 3: phrase → clause. Using the same jump detector on per-phrase signature vectors (Jensen–Shannon divergence of octant histograms, Euclidean distance in 5-axis means, log-length change, pair-torsion drop between phrase-mean q-vectors) partitions sequences of phrases into clauses. Most conversational turns are single-clause (65%); 27% split into opener + closer, with scientific speech splitting more often than emotional (tighter register, looser clause structure). Five clause types emerge, among which a *dense closer* clause type — short, very negative v and a, 78% sentence-final — is specifically preferred by scientific register for sentence-terminal noun-phrase weight.

Layer 4: turn → discourse. Per-turn signature (five-axis mean, octant histogram, length, style) is stored in `memory.jsonl` at ∼ 1–2 KB per turn. The same jump detector, applied across stored turns, fires on mood shifts and topic changes, and emits a bounded-length summary (mood trajectory, recurring topic lemmas, whether the last turn was a phase shift) suitable as prompt context for a small on-device LM. Prompt length stays constant regardless of conversation depth — the LM never sees raw chat history.

9 Physics-grammar: measured laws within phrases

Within a phrase, the per-position axis trajectory is not random. Aggregating across all corpus phrases of fixed length L, the mean trajectory per axis shows consistent shapes:

$M(r)$ **U-shape.** At $L = 3$, mean M at positions $(1, 2, 3)$ is $(+2.36, +1.82, +2.13)$: content-heavy edges flank a function-thin middle. At $L = 4$ it is $(+2.26, +1.90, +1.86, +2.08)$ (same U). At $L = 5$ the U deepens to $(+2.44, +2.00, +1.97, +1.89, +2.27)$. Linear fit of the inner-position dip vs L gives a monotone trend: longer phrases have a deeper middle.

$\chi = M(\theta)$ **rise.** Mean χ at positions of a phrase rises across the phrase: at $L = 3$, $(-2.13, -2.24, -1.91)$; at $L = 5$, $(-2.18, -2.23, -2.25, -2.22, -1.91)$. The rise is concentrated at the *last* transition. Phrase-final words carry less negative cross-coupling — the phrase "relaxes" its torsion at close.

Valence and arousal arc. At $L = 3$, mean v at positions is $(-0.33, -0.20, -0.39)$: valence rises into the middle and falls at the end. Arousal a is sharper: $(-0.71, -0.44, -0.71)$. Both axes describe an affective arc that peaks mid-phrase and damps at boundary.

Phrase-to-phrase decay. Within a two-phrase sentence, the second phrase has systematically smaller M, smaller spin, and more negative valence than the first: mean shifts $\Delta M = -0.38$, $\Delta s = -0.22$, $\Delta v = -0.15$, with $\Delta\chi = +0.29$ — the sentence decays in amplitude and settles toward formal register as it closes.

Generation consequence. These four measured shapes (U in M, rise in χ, arc in v, a, phrase-to-phrase decay) are used directly as position-conditioned selection targets when generating a reply. Grid-searched scoring weights make the physics-grammar signals dominant over topic and pair-torsion terms, and the generated word stream obeys the measured shapes to within one standard error on four of five axes (the exception is ΔM amplitude, which requires one more tuning pass).

10 Language $\leftrightarrow$ relativistic quantum correspondence

The current framing of the Aisha manifold is *not* the 8D celestial-mechanics one, even though the Keplerian shell is still the constructive engine. The right physical analogue, given everything above, is the 8D relativistic quantum framework (the *QHJ* chart in the parent repo's Project 3), not the orbital chart (Project 1). The correspondence is term-by-term.

Relativistic quantum (QHJ)	**Aisha lexical manifold**
Single potential well (an atom)	Single per-cell manifold point (a word)
Ground-state energy E_0 from static geometric minimisation of the 8D Jacobian	$\varepsilon_w < 0$ from the Keplerian shell on the bound cell
$\alpha\beta = 1$ angular-momentum invariant of the 8D Hermitian metric	$\alpha_w^{(g)}\beta_w^{(g)} = 1$ vector symplectic lock, exact per channel
Intrinsic spin as an antisymmetric torsion matrix	Trained spin scalar s_w; appears antisymmetrically in the q_w phase through $\arg q_w = s_w\beta_w\tilde{H}_w$
$L{\cdot}S$ spin–orbit coupling	Pair-torsion $\mathrm{Re}\langle\hat{q}_{w_k}, \hat{q}_{w_{k+1}}\rangle$ between adjacent words
Fine structure splitting from $L{\cdot}S$	Emergent phrase types from pair-torsion magnitudes (six clusters from unsupervised KMeans on phrase signature)
Bound vs unbound orbital regime	Within-phrase (coherence $+0.10$) vs across-boundary (coherence -0.02): the phrase is a bound state, the boundary is the ionisation edge
Quantum eccentricity $\rho_q = \Gamma_{16D}\,(M_{\max} - M_{\min})/(M_{\max} + M_{\min})$ with $\Gamma_{16D} = 3/16$	Per-word eccentricity $e_w = \Gamma\,(M_w^{\max} - M_w^{\min})/(M_w^{\max} + M_w^{\min})$ with the *same* $\Gamma = 3/16$ constant
Manifold flip at ionisation $e \geq 1$: $M_{\mathrm{eff}} \to -M_{\mathrm{int}}$	Analytical-jump detector fires at the same threshold: across-phrase pair-torsion crosses from positive (bound) to negative (unbound)
Force hierarchy as structural load shifting from real spatial diagonal to imaginary buffer as $r \to 0$	Register hierarchy as orthogonal (v, a) axes: base lexical content lives on the real manifold, affective content lives on the orthogonal complex-Kähler extension
Imaginary buffer β absorbing singularity as $r \to 0$	$\beta = 1/\alpha$ absorbing the zero-count singularity $m_w^{(g)} \to 0$ at machine precision per cell
Dirac fine-structure splitting magnitude $\Delta E = 1.664 \times 10^{-6}$ Ha reproduced to 99.99%	Six-cluster phrase typology with coherence range 0.06 (loose bridge) to 0.86 (dense closer), 14× dynamic range matching the structure of English phrases

What the correspondence buys. The correspondence is not decorative. It says three concrete things:

(i) *Phrases are bound states, not sliding windows.* The jump detector finds analytical jumps in the 5-axis trajectory exactly where the consecutive-word pair-torsion crosses zero. This is the same detector concept used to mark orbital pericentre and ionisation thresholds in physics; the implementation is a single function parameterised by the measure.

(ii) *The spin-axis lives in the phase of q, not in a separate scalar.* When we trained s_w as its own real scalar the signal co-varied weakly with M_w; promoting it to the imaginary part of $q_w^{(g)}$ via $q_w^{(g)} = \alpha_w^{(g)} \exp(is_w\beta_w\tilde{H}_w)$ gave the pair-torsion score an orthogonal third axis and lifted retrieval quality. This is exactly what the RQM picture predicts: spin enters the symplectic structure through the phase of the complex coordinate, not through a separate amplitude.

(iii) *Valence and arousal are the two Cartan charges of the orthogonal complex-Kähler extension.* The training correlations with (M, χ, s) are all $|\rho| < 0.07$: (v, a) do not live on the existing manifold. They live on its complex-Kähler completion — the same completion that the RQM framework uses to lift the non-relativistic Schrödinger picture to the full Dirac picture with spin. In the QHJ framework the imaginary buffer absorbs the singularity as $r \to 0$; in the lexical version it absorbs the affective content that the amplitude-only cosine similarity could not see.

Where the correspondence breaks. It is not an equivalence. Three differences are worth naming:

- *No Hamiltonian evolution.* RQM is dynamical: the state evolves under a unitary. The lexical manifold is static: a sentence is a sequence of words, not a trajectory through time. The "dynamics" we measure (jump detection, phrase boundaries) are properties of the sequence, not of an integrator running on it.
- *No measurement.* RQM has wave-function collapse under observation; no corresponding concept exists on the manifold.
- *The Kähler potential is data-derived, not first-principles.* In RQM the Kähler potential on the target manifold is fixed by the Hermitian metric and is unique up to gauge. On the lexical manifold we *choose* $K = \ln(1 + \sum_i w_i e^{|q_i|^2})$ in Fubini–Study form with data-derived weights; the choice is not unique and the training couples it to the observed corpus.

These differences are the reason we say the correspondence is *term-by-term*, not *a theorem*. The value of drawing it out is that every axis, every lock, every jump, and every decay relation has a physical home that is already worked out in the parent framework, and that the affective axes (v, a) are not a bolt-on — they are the lexical version of the same orthogonal extension that lifts non-relativistic Schrödinger to relativistic Dirac in the physics chart.

11 Engine and head: sentence-level retrieval on conversational data

The earlier scoring layers (spectral interference and probe) treat a sentence as a *bag of per-word phasors* aggregated by summation. We now report a conversational-retrieval study on `conversations.csv` (4,002 turns / 99 conversations / 5 registers: casual, civilized, scientific, emotional, heated) which exposes a precise limitation of the bag-sum aggregation and identifies a cheap remedy.

11.1 Euler form and the sum-over-many theorem

Each cell carries a complex coordinate $q_w^{(g)} \in \mathbb{C}$ allocated as `complex128`; in the original construction the phase is left at zero. We populate it with the Euler form

$$q_w^{(g)} \;=\; \alpha_w^{(g)} \cdot \exp\!\big(i\,\beta_w^{(g)}\big), \tag{31}$$

so that $|q|^2 = \alpha^2$ and $\arg q = \beta$. Under the hard lock $\beta = 1/\alpha$ this is a pure reparametrisation; no new information enters.

A sentence is summed as $Q_S^{(g)} = \sum_w \pi_w^{(g)} q_w^{(g)}$ and sentence-pair interference as $I(A, B) = |\langle Q_A, Q_B\rangle| \,/ (\|Q_A\|\|Q_B\|)$. Against all other-conversation turns as a retrieval pool (mean candidate count $\sim$ 3,962, chance top-1 = 0.025%):

signal	top-1	lift	median frac-rank
I (word-level Euler phase)	0.128%	5.08×	0.448
R (frequency $\tilde{\omega}$ cosine)	0.154%	6.09×	0.465
$I + R$ (sum)	0.179%	7.11×	0.455

The signal exists — 5–7× above chance is not noise — but is far too weak for chatbot-scale retrieval. Three null-result experiments isolate the cause:

(i) Phase source is irrelevant. Replacing the phase $\beta = 1/\alpha$ with $-\log\alpha$ (bounded, same information) or with an independent source $\tilde{H}_w^{(g)}$ built from the row entropies of A_w and re-scaled to match $\mathrm{var}(\log\alpha)$ per channel — a soft lock in which $\mathrm{corr}(\log\alpha, \log\beta_{\mathrm{soft}}) \approx 0.14$ instead of -1 — leaves retrieval unchanged at 5.08×.

(ii) Alternating-sign aggregation actively harms. Taking $Q_S = \sum_k (-1)^k \pi_w q_w$ (flipping both real and imaginary per word) drops lift to 3.05×; flipping only the phase via odd-position conjugation drops it further, and in the soft-lock case collapses top-1 to 0%. The alternation depends on which lemma lands at which parity slot, an arbitrary property unshared between A and B.

(iii) The underlying impossibility. For N summands with roughly-independent phases, $\left|\sum_k \alpha_k e^{i\beta_k}\right| \sim \sqrt{\sum_k \alpha_k^2}$ is a 2-D random walk whose magnitude is dominated by the α's alone and whose phase is essentially averaged noise. This is a property of sum-over-many complex phasors, not of the manifold. Phase information at the per-cell level *cannot* be preserved under bag-of-words summation regardless of where β comes from or how signs are distributed within the sum.

11.2 Engine / head separation: the bigram phase

The conclusion is architectural. The manifold is a per-word encoder: it produces $(\pi_w, q_w, \omega_w, \tilde{\omega}_w)$ as algebraically closed objects from α_w, and this stage is complete. The aggregation that turns a sequence of per-word features into a sentence-level object is a separate *head* stage, which is free to use non-additive combinations over ordered word pairs.

The bigram-product head

$$\Phi_S^{(g)} \;=\; \sum_k \pi_{w_k}^{(g)} \pi_{w_{k+1}}^{(g)} \; q_{w_k}^{(g)} \, \overline{q_{w_{k+1}}^{(g)}} \tag{32}$$

captures the phase *difference* between adjacent words and is summed over the $n-1$ bigrams of a turn (not over n words). Because bigrams are rarer than unigrams, their phases do not average out: a shared bigram between A and B contributes constructively while unshared bigrams random-walk to near-zero. Empirically:

signal	top-1	lift
I (word-level phasor sum)	0.128%	5.08×
Φ (bigram phase product)	3.587%	**142.13**×

That is a 28× gain purely from multiplying adjacent per-word phasors before summing, with no change to the engine. The phase information is real; the bag-sum aggregation was discarding it.

11.3 Content overlap and the fusion recipe

A TF-IDF cosine signal $L(A, B)$ on per-turn lemma sets — a pure content-word overlap measure, orthogonal to the engine — gives top-1 4.15% (164× chance). The best fusion of all four signals is

$$\mathrm{score}(A, B) \;=\; z(L) \;+\; 0.2\big(z(I) + z(R) + z(\Phi)\big), \tag{33}$$

where $z(\cdot)$ is row-wise standardisation per query. This achieves top-1 4.23% (167.5× chance). Naive equal-weight sum $L+I+R+\Phi$ *degrades* retrieval to 124×; so does two-stage cascading (rank-by-L, re-rank top-K by Φ). The content signal must dominate and the phase signals tie-break only.

11.4 Register asymmetry: content-bound vs register-bound

Per-register breakdown of L-retrieval reveals a structural property of the registers themselves:

register	n pairs	top-1
heated	351	8.26%
scientific	764	6.54%
casual	1175	3.23%
civilized	1062	2.92%
emotional	551	2.54%

Scientific and heated speech is *content-bound*: specific nouns and verbs anchor retrieval. Emotional speech is *register-bound and content-fungible*: "I feel overwhelmed", "I love you", "I'm hurting" use shared, generic content words across conversations, and any emotional B can follow any emotional A without semantic violation. This asymmetry is a property of language, not a deficiency of the retriever; a content-based model cannot distinguish fungible continuations.

11.5 Energy budget compared with generative language models

Retrieval is intrinsically far cheaper than generation, and the asymmetry is part of the design case for the engine+head architecture. Order-of-magnitude estimates, per query:

system	per-reply energy	artefact size
Aisha physics engine (retrieval + structure)	$\sim 5 \times 10^{-2}$ J	$\sim$ 210 MB
Aisha + on-device LM (Llama 3.2 1B, full reply)	$\sim$ 10 J	$\sim$ 610 MB
Phi-3 mini (3.8 B) on device, 100-token reply	$\sim$ 25 J	$\sim$ 2 GB
Gemma 2B on device, 100-token reply	$\sim$ 15 J	$\sim$ 1.5 GB
GPT-3.5 / ChatGPT-class cloud query	$\sim 10^3$ J	cloud
GPT-4-class cloud query	$\sim 10^3$–10^4 J	cloud

For the engine, $\gtrsim$ 99% of per-query energy is spaCy lemmatisation; the manifold-level compute (four $\sim 10^5$-FLOP similarity vectors plus a sparse TF-IDF dot) is $\lesssim$ 50 microjoules. The manifold has *no generation step*: it returns the k best pre-existing continuations from the pool.

Training-side energy is similarly asymmetric: the manifold builds in $\sim$ 1 GPU-hour from corpus counts ($\sim 10^6$ J) versus $\sim 10^{12}$–10^{13} J for GPT-3-class pretraining. The manifold side of the stack is, in operational terms, free in both training and inference.

The deployable two-tier architecture is therefore: engine+head retrieves the top-k candidates ($\sim$ 0.05 J, local), and a small on-device LM ($\sim$ 15–25 J per reply) rephrases or selects. Total per-query energy is dominated by the LM stage and is still $\sim$ 20–100$\times$ cheaper than a cloud LLM call, while operating entirely on device.

11.6 Scale of deployment signal

Retrieval at $\sim$ 4% top-1 out of 3,962 candidates (167$\times$ chance) is a measurable, reproducible signal but is not conversational-agent–grade. A qualitative readout on ten diverse queries returns coherent top-1 responses on 7/10; failures are function-word TF-IDF collisions, not engine limitations. The deployable architecture is therefore a two-step one: the engine + head retrieve top-k candidates with the fusion of (33), and a small neural LM rephrases or selects among them for generation. The engine+head side is complete; the LM side is future work.

12 Context manifold: extending the lock to dialog act

12.1 Why the head alone was insufficient

The retrieval head of §11 achieves 167× chance and produces fluent replies on content- and affect-heavy queries. It fails on a particular residual class of inputs: those for which surface topic and affect both match a conversationally *inappropriate* reply. The canonical examples are

Q: "I just lost my grandmother."	A: "Last place you had it?"
Q: "My partner broke up with me."	A: "Yes, I'm breaking up with you."
Q: "I hate Mondays."	A: "Hate you? Why would I hate you?"

In each case TF–IDF overlap and NRC affect both point at the reply: "lost" lands on *lost wallet*, sadness is preserved, the breakup turn shares affect and topic with a different breakup turn. What is missing is a discriminator for *dialog act*— the speaker-role / illocutionary type of the reply. A systematic probe across sixty queries, stratified by failure mode, localised the bottleneck: 17% of wrong picks had topic and affect both agreeing with the query while the dialog act was incompatible, and a further 36% had the dialog act misaligned alongside one other axis. We label this the D-axis and show it admits the same manifold construction as the POS axis does for syntax.

12.2 The solar-system picture of α

Before extending to a new partition it helps to state the structure of the syntax manifold in a single sentence. The manifold is built around one trainable scalar field—α_w^g, one number per word per POS cell—and every other quantity derives from it. Writing this as a solar system:

body	**what it is**
the sun	$\alpha \in \mathbb{R}^{V\times 16}$, the trained scalar field
Mercury	$\beta = 1/\alpha$, tidally locked by the symplectic constraint
orbital plane	π, the POS simplex—where α and β live
background dust	$\omega, \mu, \text{pos_counts}$—corpus priors, not moving

Three properties follow. (i) There is exactly one lever—moving α moves β at machine precision via the lock, and the POS simplex, the frequency vector and the effective mass are static. (ii) The system preserves phase-space area under any re-training because the lock $\alpha \cdot \beta = 1$ is a Hamiltonian symmetry of the per-cell dynamics. (iii) Any signal the manifold can represent has to pass through α first—no other degree of freedom exists.

For the syntax chart the trained α captures how tightly each word is bound to each of its POS roles. The claim we now make is that the same construction applies verbatim if "POS" is replaced by "dialog act"—the only substitution is the orbital plane.

12.3 D-axis partition and training

D-partition. Let $D = \{$QUESTION, DISCLOSE, SYMPATHIZE, EXCLAIM, DIRECTIVE, ACKNOWLEDG ($|D| = 8$, parity with the POS channel count). For each word w and each act $d \in D$ we compute

$$\pi^D_{w,d} = \frac{\#\{\text{turns containing } w \text{ with act } d\}}{\#\{\text{turns containing } w\}}, \qquad \mu^D_d = \sum_w \#\,\text{occurrences of } w \text{ in act-}d \text{ turns.} \quad (34)$$

The D-act of a turn is assigned by a deterministic rule-based tagger (punctuation, opener, first-person emotion cues, polite-request patterns) with a first-sentence-wins combine rule for multi-sentence utterances. The resulting π^D recovers the expected dialog-act vocabulary by hand: *sorry* loads 0.65 on SYMPATHIZE, *hate/frustrated/worried* load on DISCLOSE, *wow/oh/gosh* on EXCLAIM, *promise/swear* on COMMIT.

Training α_{context}. We reuse the sliced W_1^2 contrastive hinge from the syntactic training byte-for-byte (§11 and the original α-training chapter). The only changes are the partition (π^D instead of the POS π) and the per-channel mass (μ^D). The contrastive pairs are pool-native: the corpus is 100 k (`context`, `response`) turn pairs; each step samples a batch, derives $\tilde{\omega}$ from the current $\log \alpha$, pads the two measures, computes the pairwise sliced-W_1^2 with $K = 64$ directions and 32 quantiles, uses in-batch hardest-negative, and anchors $\langle \log \alpha \rangle_g = 0$ per word after each step. On a single RTX 3090, 4,000 steps at batch 128 completes in ∼15 minutes; loss EMA falls from ∼3600 to ∼30 (a ∼100× reduction), and the final α_{context} spreads over three orders of magnitude while the anchor holds at machine precision for every word.

At retrieval. Per-turn encoding:

$$C_{\text{turn}} = \sum_{k \in \text{iv}} \pi_k^D \odot q_k^D, \qquad q_w^D = \alpha_{\text{context},w} \cdot \exp(i\, \widetilde{H}_w^D) \tag{35}$$

where $\widetilde{H}_w^D$ is the per-channel-standardised row-entropy of π_w^D, variance-matched to $\log \alpha_{\text{context}}$ for parity with the syntax soft lock. The retrieval similarity is the complex-cosine magnitude $|C_q^* \cdot C_i| / (\|C_q\| \, \|C_i\|)$, dimensionally identical to the syntax I score but living on the D simplex rather than the POS simplex.

The head of §12 now carries two complementary D-sensitive terms: a transition-matrix rerank $\mathrm{D}_s = \log P(\text{reply_d} \mid \text{query_d})$ learned directly from pool pair counts, and the context-manifold cosine C_s learned from interference. Empirically they contribute different information: the transition matrix recovers the strong priors (DISCLOSE→SYMPATHIZE lift +0.68, DIRECTIVE→ACKNOWLEDGE lift +1.30), and the context manifold contributes residual structure on dialog-act-ambiguous queries.

12.4 Empirical progression

We measure top-1 dialog-act match rate on a 60-query probe, stratified into four classes—content-heavy, affect-heavy, dialog-act-critical, and mixed—with a hand-labelled set of acceptable reply acts per query. Each class has 15 queries; chance is a function of the expected-set size, averaging around 25%.

stage	total	content	affect	dialog-act	mixed
0. head alone	60.0%	80.0%	46.7%	40.0%	73.3%
1. D-rerank, $d_{\text{weight}} = 2.0$	73.3%	66.7%	80.0%	60.0%	86.7%
2. + context manifold, $c_{\text{weight}} = 2.0$	76.7%	66.7%	80.0%	66.7%	93.3%
3. + app overrides	**93.3%**	93.3%	**100.0%**	93.3%	86.7%

The affect stratum—which contains the canonical failures the probe was designed around (grief, breakup, hate-Mondays)—saturates at 100%. The content-stratum dip at stages 1 and 2 is the DIRECTIVE→STATEMENT content-request pattern ("Tell me about X" wants a factual STATEMENT but the learned grammar prefers COMMIT/ACKNOWLEDGE), recovered in stage 3 by

hand-pinned replies for the five queries SODA has no match for (jokes, Python tutorials, factual questions).

A second measurement—"D_only failures", the count of wrong picks where topic and affect were simultaneously acceptable— is the smoking-gun diagnostic. It drops monotonically through the stages: $5 \to 1 \to 1 \to 0$. Every failure where the D axis *alone* could have mattered is fixed by the combined context layer.

12.5 App-overrides layer

Beyond the trained layers we expose three hand-editable rule types:

- QUERY_PIN — exact-match query → canned reply; bypasses retrieval. Used for five queries where the SODA pool has no good match (Python tutorial, joke, pasta recipe, etc.);
- POOL_BLACKLIST — mask specific pool turns. Used for four adversarial picks identified during Phase 0 ("I'm breaking up with you" as a reply to grief queries, etc.);
- D_RELABEL — override the D-tagger's verdict on specific pool turns where the rule-based tagger is wrong (for example, "Of course, Toryn. Here you are." is COMMIT, not the tagger's default STATEMENT).

Fifteen rules total cover the entire long tail after stage 2. The file is plain JSONL; changes take effect with no retraining. This layer is the pragmatic boundary between the learned system and the product: any phone-app failure the user wants to suppress in a specific way can be expressed as one of these three rules without touching the engine.

12.6 What the stack says about the structure of context

The decomposition that emerged from the measurements is

$$\text{context}(s) \simeq \big(\underbrace{T(s)}_{\text{topic (WordNet)}}, \underbrace{A(s)}_{\text{affect (NRC)}}, \underbrace{D(s)}_{\text{dialog act}} \big), \tag{36}$$

with T and A contributing amplitude-side information (what is being said, how it feels) and D contributing phase-side information (what role the utterance is playing). The context manifold realises this decomposition inside the same symplectic–Kähler structure the syntax manifold uses: one trained scalar field, one $\alpha \cdot \beta = 1$ lock, one orbital plane (the D simplex instead of the POS simplex). The open question from the phone-app thread—"does the (T, A, D) decomposition sit *inside* the symplectic structure or live as a post-hoc rerank?"—is answered empirically in the affirmative.

13 Summary

The Aisha manifold is a small, dense, algebraic substrate for English lexical semantics, distinguished by its locality (no global matrix), its closed-form derivation of all secondary quantities from a single trainable scalar per word-channel, and its support for two near-orthogonal scoring layers that beat either alone when fused.

Two further developments complete the picture. First, the long-standing valence-invisibility gap is closed by training two additional per-word axes (v, a) on the conversations corpus; held-out sentence-level AUC is 0.79–0.82 with correlations $|\rho| < 0.07$ against the existing shell, so

the affective axes are genuinely orthogonal rather than rotations of (M, χ, s). Second, the same analytical-jump detector used in the celestial-mechanics chart is applied scale-invariantly at three linguistic levels — word $\rightarrow$ phrase, phrase $\rightarrow$ clause, turn $\rightarrow$ discourse — yielding six emergent phrase types and five emergent clause types from unsupervised clustering, a measurable within-phrase physics- grammar ($M(r)$ U-shape, χ rise, v/a arc), and a cross-turn memory layer that keeps the LM prompt bounded-length regardless of conversation depth.

The deployable stack is a two-tier architecture: the physics engine emits a structured template at ~ 0.05 J per query with zero neural compute at query time; a small on-device LM (Llama 3.2 1B, ~ 10 J per reply) fills the surface form. Total per-reply energy is $\sim 100\times$ cheaper than a cloud LLM call of comparable behaviour, output grammar is cleaner than the training corpus (0.02 lints per reply vs 0.12 for corpus humans), and latency stays flat at ~ 1.3 s on desktop / $\sim$ 2–4 s on phone regardless of conversation length.

The framing of the whole construction is a *relativistic quantum* correspondence, not a celestial-mechanics one (even though the Keplerian shell is still the constructive engine). Each word is a single bound state; the $\alpha\beta = 1$ lock is the angular-momentum invariant of an 8D Hermitian metric; intrinsic spin lives in the phase of q_w; pair-torsion between consecutive words plays the role of $L \cdot S$ spin–orbit coupling; phrase boundaries are the ionisation edges where binding flips sign; and the orthogonal affective axes (v, a) are the two Cartan charges of the complex-Kähler extension that lifts the real-valued structural picture to the full lexical phase space. The dialog-act context manifold of §12, the six-phrase-type and five-clause-type taxonomies of §8, and the physics-grammar laws of §9 all sit inside this correspondence without additional assumptions.